

Complete Particle Spectrum: From Standard Model Complexity to T0 Universal Field

Contents

.1	Introduction: The Complete Particle Count	2
	Standard Model Particle Inventory	2
	T0 Theory Universal Field Approach	2
.2	Complete Standard Model Particle Catalog	3
	Generation Structure	3
	Gauge Bosons and Higgs	3
	Antiparticles	3
	Composite Particles	3
.3	Hypothetical and Dark Sector Particles	4
	Beyond Standard Model Candidates	4
	Supersymmetric Particles	4
.4	T0 Theory: Universal Field Unification	5
	The Basic Idea	5
	Complete T0 Particle Spectrum	5
	Neutrinos as a Limiting Case	5
	Antiparticles: Sign of the Excitation	6
.5	Comprehensive Comparison	6
	Particle Count Comparison	6
	Explanatory Power Comparison	6
.6	Experimental Implications	6
	Testable T0 Predictions	6
	Lepton Corrections	7
	Neutrino Mass Ratios	7
	Continuous Particle Spectrum	7
	Dark Sector Predictions	7
	Dark Matter as Sub-Threshold Excitations	7
	Axion-Like Particles	8
.7	Proposed Answers to Open Questions in Particle Physics	8
	The Generation Problem	8
	The Hierarchy Problem	8
	The Strong CP Problem	8
.8	Cosmological and Astrophysical Implications [S]	8
	Big Bang as Universal Field Excitation	9
	Stellar Nucleosynthesis	9
	Black Holes and the Information Paradox	9
.9	Philosophical Implications	9
	From Particles to Excitation Patterns	9
	Unity in Diversity	9
	The Question of Consciousness	9
.10	Conclusion	10
	Result	10

The Picture in Words	10
Context	10

Abstract

This analysis presents the spectrum of known particles in both the Standard Model and the T0 theory framework. While the Standard Model requires 17 fundamental particles plus their antiparticles (29 fundamental entities in total, since only the 12 fermions and the W have distinct antiparticles) and hundreds of composite particles, the T0 theory describes how all particles arise as different excitation strengths ε in a single universal field $\delta m(x, t)$. We provide detailed mappings of each particle type, from leptons and quarks to gauge bosons and hypothetical particles such as axions and gravitons, and show how the T0 framework brings these particles together in one framework through the universal equation $\mathcal{L} = \varepsilon \cdot (\partial \delta m)^2$ with the single parameter $\xi = 1.33 \times 10^{-4}$ (plus integer structural choices as motivated settings and SI anchors for unit conversion).

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. This applies in particular to the cosmic sector, which the current corpus deliberately sets aside, since the Standard Model does not derive it either (P39).

.1 Introduction: The Complete Particle Count

Standard Model Particle Inventory

The Standard Model of particle physics is the experimentally best-confirmed theory of fundamental particles and forces; its particle spectrum and parameters, however, are largely fixed empirically rather than derived. The complete inventory comprises:

Standard Model: Particles and Parameters

Fundamental Particles: 17 types

- 6 Leptons (Electron, Muon, Tau + 3 Neutrinos)
- 6 Quarks (up, down, charm, strange, top, bottom)
- 4 Gauge Bosons (Photon, $W^\pm$, Z^0 , Gluon)
- 1 Higgs Boson

Antiparticles: distinct antiparticles only for the 12 fermions and the W ; photon, gluon, Z^0 and Higgs are their own antiparticles

Composite Particles: 100+ Hadrons, Mesons, Baryons

Total Known Particles: 200+ different entities

Free Parameters: 19+ experimentally determined values

T0 Theory Universal Field Approach

The T0 theory proposes an alternative description: all particles as excitations of a single field:

T0 Universal Field Simplification

One universal field: $\delta m(x, t)$

One universal equation: $\mathcal{L} = \varepsilon \cdot (\partial \delta m)^2$

One universal parameter: $\xi = 1.33 \times 10^{-4}$ (plus motivated structural choices)

Infinite particle spectrum: Continuous ε values

Automatic antiparticles: $-\delta m$ (negative excitations)

Common description: From photons to Higgs bosons

.2 Complete Standard Model Particle Catalog

Generation Structure

The Standard Model organizes fermions into three generations:

Generation	1st	2nd	3rd
Leptons	e^- (0.511 MeV) ν_e (< 2 eV)	μ^- (105.7 MeV) ν_μ (< 0.19 MeV)	τ^- (1777 MeV) ν_τ (< 18.2 MeV)
Quarks	u (+2/3, 2.2 MeV) d (-1/3, 4.7 MeV)	c (+2/3, 1.3 GeV) s (-1/3, 95 MeV)	t (+2/3, 173 GeV) b (-1/3, 4.2 GeV)

Table 1: Standard Model Three-Generation Structure

Gauge Bosons and Higgs

Particle	Symbol	Mass	Charge	Force
Photon	γ	0	0	Electromagnetic
W Boson	$W^\pm$	80.4 GeV	± 1	Weak (charged)
Z Boson	Z^0	91.2 GeV	0	Weak (neutral)
Gluon	g	0	0	Strong
Higgs	H^0	125 GeV	0	Mass generation

Table 2: Standard Model Gauge Bosons and Higgs Boson

Antiparticles

Each fermion has a corresponding antiparticle:

- **Antileptons:** e^+ , μ^+ , τ^+ , $\bar{\nu}_e$, $\bar{\nu}_\mu$, $\bar{\nu}_\tau$
- **Antiquarks:** $\bar{u}$, $\bar{d}$, $\bar{c}$, $\bar{s}$, $\bar{t}$, $\bar{b}$
- **Self-conjugate Bosons:** γ , Z^0 , g , H^0 (their own antiparticles)

Total fundamental particles: 17 particles + 12 distinct antiparticles = **29 fundamental entities**

Composite Particles

Quarks combine to form hundreds of composite particles:

Baryons (3 quarks):

- Proton: uud (938.3 MeV)
- Neutron: udd (939.6 MeV)
- Lambda: uds (1115.7 MeV)
- Sigma particles: Σ^+ (uus), Σ^0 (uds), Σ^- (dds)
- Xi particles: Ξ^0 (uss), Ξ^- (dss)
- Omega: Ω^- (sss)
- Charm baryons: Λ_c^+ , Σ_c , etc.
- Bottom baryons: Λ_b^0 , Σ_b , etc.

Mesons (quark-antiquark pairs):

- Pions: π^+ ($u\bar{d}$), π^0 ($u\bar{u} - d\bar{d}$), π^- ($d\bar{u}$)
- Kaons: K^+ ($u\bar{s}$), K^0 ($d\bar{s}$), K^- ($s\bar{u}$), $\bar{K}^0$ ($s\bar{d}$)
- Eta particles: η , η'
- Rho mesons: ρ^+ , ρ^0 , ρ^-
- J/psi: $c\bar{c}$ (charm-anticharm)
- Upsilon: $b\bar{b}$ (bottom-antibottom)

Total composite particles: Over 200 experimentally observed hadrons

.3 Hypothetical and Dark Sector Particles

Beyond Standard Model Candidates

Particle	Mass Range	Purpose	Status
Graviton	0	Quantum gravity	Hypothetical
Axion	$10^{-6} - 10^{-3}$ eV	Dark matter	Hypothetical
Sterile Neutrino	eV - keV	Neutrino anomalies	Disputed
Dark Photon	MeV - GeV	Dark sector	Hypothetical
WIMP	GeV - TeV	Dark matter	Hypothetical
Magnetic Monopole	10^{16} GeV	GUT theories	Hypothetical

Table 3: Hypothetical Particles Beyond the Standard Model

Supersymmetric Particles

Supersymmetry (SUSY) predicts partner particles for each Standard Model particle:

Sparticles (supersymmetric partners):

- **Sleptons:** $\tilde{e}$, $\tilde{\mu}$, $\tilde{\tau}$, $\tilde{\nu}_e$, $\tilde{\nu}_\mu$, $\tilde{\nu}_\tau$
- **Squarks:** $\tilde{u}$, $\tilde{d}$, $\tilde{c}$, $\tilde{s}$, $\tilde{t}$, $\tilde{b}$
- **Gauginos:** $\tilde{\gamma}$ (Photino), $\tilde{W}$ (Wino), $\tilde{Z}$ (Zino), $\tilde{g}$ (Gluino)
- **Higgsinos:** $\tilde{H}^0$, $\tilde{H}^\pm$

Total SUSY particles: 100+ additional hypothetical particles

Current status: No SUSY particles discovered despite extensive LHC searches

.4 T0 Theory: Universal Field Unification

The Basic Idea

The T0 theory describes all particles as different excitation strengths in the same field:

$$\boxed{\text{All particles} = \text{Different } \varepsilon \text{ values in } \delta m(x, t)} \quad (1)$$

where $\varepsilon = \xi \cdot E^2$ with the universal scale parameter $\xi = 1.33 \times 10^{-4}$.

Complete T0 Particle Spectrum

Particle Type	Examples	ε Range	T0 Interpretation	SM Comparison
Massless Bosons	Photon (γ)	$\varepsilon \rightarrow 0$	Limiting case of the field	Gauge boson
Ultra-light Particles	Axions, dark photons	$10^{-16} - 10^{-10}$	Sub-threshold excitations	Dark matter candidates
Neutrinos	ν_e, ν_μ, ν_τ	$10^{-12} - 10^{-7}$	Minimal field excitations	Separate neutrino fields
Light Leptons	Electron (e^-)	$\sim 3 \times 10^{-8}$	Weak field excitation	Charged lepton
Light Quarks	Up (u), Down (d)	$10^{-6} - 10^{-5}$	Confined excitations	Color-charged quarks
Medium Leptons	Muon (μ^-)	$\sim 1.5 \times 10^{-3}$	Medium field excitation	Heavy lepton
Strange Particles	Strange (s), Charm (c)	$10^{-3} - 10^{-1}$	Medium-strong excitations	2nd generation quarks
Heavy Leptons	Tau (τ^-)	~ 0.42	Strong field excitation	Heaviest lepton
Heavy Quarks	Top (t), Bottom (b)	$1 - 10$	Very strong excitations	3rd generation quarks
Weak Bosons	$W^\pm, Z^0$	$860 - 1100$	Electroweak scale excitations	Gauge bosons
Higgs Sector	Higgs (H^0)	~ 2080	Structural foundation	Scalar field

Table 4: Complete Particle Spectrum in the T0 Theory

$\varepsilon = \xi E^2$ is dimensionful, so the table values depend on the unit. With $\xi = 1.33 \times 10^{-4}$ one obtains in MeV^2 : electron 3.5×10^{-5} , muon 1.48, tau 420; in GeV^2 : 3.5×10^{-11} , 1.5×10^{-6} , 4.2×10^{-4} . The lepton entries of the table (3×10^{-8} , 1.5×10^{-3} , 0.42) arise only with mixed units ($\xi \cdot m[\text{GeV}] \cdot m[\text{MeV}]$). In the same mixed units the W (80.4 GeV) gives 860 (in GeV^2 : 0.86), the Z gives 1106 (1.11 GeV^2) and the Higgs (125.1 GeV) gives 2080 (2.08 GeV^2). The ultra-light row follows from the axion section.

Neutrinos as a Limiting Case

Neutrinos deserve special attention because they represent the transition from particles to the vacuum:

$$\begin{aligned}
\nu_e : \quad \varepsilon_1 &\approx 10^{-12} \quad (m_1 \sim 0.0001 \text{ eV}) \\
\nu_\mu : \quad \varepsilon_2 &\approx 10^{-8} \quad (m_2 \sim 0.009 \text{ eV}) \\
\nu_\tau : \quad \varepsilon_3 &\approx 3 \times 10^{-7} \quad (m_3 \sim 0.05 \text{ eV})
\end{aligned} \tag{2}$$

Physical interpretation: Neutrinos are ghostly because their field excitations are so weak that they barely interact with matter. They represent the boundary between detectable particles and the vacuum state.

Antiparticles: Sign of the Excitation

In the T0 theory, antiparticles require no separate treatment:

$$\boxed{\text{Antiparticle} = -\delta m(x, t)} \tag{3}$$

Examples:

$$\text{Electron} : \quad \delta m_e(x, t) = +A_e \cdot f_e(x, t) \tag{4}$$

$$\text{Positron} : \quad \delta m_{e^+}(x, t) = -A_e \cdot f_e(x, t) \tag{5}$$

$$\text{Annihilation} : \quad \delta m_e + \delta m_{e^+} = 0 \tag{6}$$

In the T0 framework this removes the need for separate bookkeeping of antiparticles; they are described by the sign of the same excitation.

.5 Comprehensive Comparison

Particle Count Comparison

Category	Standard Model	T0 Theory
Fundamental Particles	17	1 Field
Antiparticles	12 fermions + w , in the same field	Same field (negative)
Free Parameters	19+	1 (ξ) + settings
Composite Particles	200+ cataloged	Infinite spectrum
Hypothetical Particles	100+ (SUSY, etc.)	Natural extensions
Dark Sector	Separate particles	Sub-threshold excitations
Gravitons	Not included	Emergent from $T \cdot m = 1$
Total Complexity	Hundreds of entities	One universal field

Table 5: Comprehensive Complexity Comparison

In the Standard Model only the 12 fermions and the w have distinct antiparticles, and they are described by the same fields as the particles; there are no separate antiparticle fields.

Explanatory Power Comparison

.6 Experimental Implications

Testable T0 Predictions

In this early version, the T0 universal field theory makes specific predictions that distinguish it from the Standard Model:

Phenomenon	Standard Model	T0 Theory
Particle Masses	17+ independent measurements	Single parameter ξ
Generation Structure	Fixed empirically	ε hierarchy
Neutrino Oscillations	Complex mixing matrices	Field interference patterns
Dark Matter	Unknown new particles	Sub-threshold excitations
Matter-Antimatter Asymmetry	Unsolved problem	Natural ξ asymmetry
Gravitation	Not part of the SM	Included in the approach
Quantum Mechanics	Probabilistic framework	Deterministic field evolution
Particle Creation/Annihilation	Complex QFT processes	Simple field dynamics

Table 6: Explanatory Power Comparison

Lepton Corrections

A contribution equal for all leptons,

$$a_t^{(T0)} = \frac{\xi}{2\pi} \times \frac{1}{12} \approx 1.77 \times 10^{-6}, \quad (7)$$

is calculated correctly but does not hold, because it is excluded by the measurement of a_e , which agrees with the Standard Model to about 10^{-12} . Authoritative is the ratio line (Doc. 018 Rev. 12, 158, 382). The deviation between the measured a_μ and the Standard Model prediction as it stood in 2021 is no longer significant with the Fermilab final result and the 2025 White Paper of the Muon $g - 2$ Theory Initiative, and therefore serves neither as evidence nor as the target of a T0 contribution; the current status and assessment are given in Doc. 382.

Neutrino Mass Ratios

$$\frac{m_3}{m_2} = \sqrt{\frac{\varepsilon_3}{\varepsilon_2}} \approx 5.5, \quad \frac{m_2}{m_1} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \approx 100 \quad (8)$$

with the values from Eq. (2): $\sqrt{3 \times 10^{-7}/10^{-8}} = 5.5$, $\sqrt{10^{-8}/10^{-12}} = 100$; the masses given there correspondingly yield $0.05/0.009 = 5.6$ and $0.009/0.0001 = 90$.

Continuous Particle Spectrum

The T0 theory predicts a continuous spectrum of particle-like excitations:

- Search for particles with ε values between known particles
- Search for missing particles in the continuous spectrum
- Test whether new particles fit the universal $\varepsilon = \xi \cdot E^2$ relation

Dark Sector Predictions

Dark Matter as Sub-Threshold Excitations

$$\delta m_{\text{dark}} = \xi \cdot \rho_0 \cdot \sin(\omega_{\text{dark}} t + \phi_{\text{random}}) \quad (9)$$

where $\varepsilon_{\text{dark}} \ll 10^{-12}$ (below the neutrino threshold).

Axion-Like Particles

In this picture, ultra-light axions would arise as **[S]**:

$$\varepsilon_{\text{Axion}} \approx 10^{-16} \text{ to } 10^{-10} \quad (10)$$

corresponding to masses $m_a \sim 10^{-6} \text{ to } 10^{-3} \text{ eV}$, with $\varepsilon \propto m^2$ and $\varepsilon_1 \approx 10^{-12}$ at 10^{-4} eV from Eq. (2): $10^{-12} \cdot (10^{-6}/10^{-4})^2 = 10^{-16}$, $10^{-12} \cdot (10^{-3}/10^{-4})^2 = 10^{-10}$.

.7 Proposed Answers to Open Questions in Particle Physics

The Generation Problem

Standard Model puzzle: Why exactly three generations of fermions?

T0 proposal: Three generations correspond to three scales in the ε spectrum (in the mixed units of Table 4):

$$\text{1st generation : } \varepsilon \sim 10^{-8} \text{ to } 10^{-6} \quad (\text{stable matter}) \quad (11)$$

$$\text{2nd generation : } \varepsilon \sim 10^{-3} \text{ to } 10^{-1} \quad (\text{medium instability}) \quad (12)$$

$$\text{3rd generation : } \varepsilon \sim 1 \text{ to } 10 \quad (\text{high instability}) \quad (13)$$

The Hierarchy Problem

Standard Model puzzle: Why is the Higgs mass so much smaller than the Planck mass?

T0 proposal [S]: The Higgs represents the structural foundation with:

$$\varepsilon_H = \xi^{-1} \approx 7500 \quad (14)$$

In this interpretation, this is the scale where the field transitions from particle-like to structure-like behavior. This value does not, however, follow from $\varepsilon = \xi E^2$: for the Higgs (125.1 GeV) one obtains 2.08 GeV^2 , or 2080 in the mixed units of Table 4.

The Strong CP Problem

Standard Model puzzle: Why is the strong CP phase so small?

T0 approach: Setting the CP phase by the universal scale from field asymmetry,

$$\theta_{CP} \approx \xi \sim 10^{-4}, \quad (15)$$

does not hold, because it lies about a factor 10^6 above the bound $\bar{\theta} \lesssim 10^{-10}$ from the electric dipole moment of the neutron; it is therefore excluded.

.8 Cosmological and Astrophysical Implications [S]

The statements in this section are speculative. The current corpus deliberately sets the cosmic sector aside, since the Standard Model does not derive it either (P39).

Big Bang as Universal Field Excitation

In this picture, the Big Bang would become a sudden excitation of the universal field:

$$\delta m(x, t = 0) = \delta m_0 \cdot \delta^3(x) \cdot e^{-H_0 t} \quad (16)$$

Particle creation would arise from this initial field excitation, with a slight asymmetry $\propto \xi$ favoring matter over antimatter.

Stellar Nucleosynthesis

Nuclear reactions can be described as field excitation transformations:

$$\delta m_{\text{light}} + \delta m_{\text{light}} \rightarrow \delta m_{\text{heavy}} + \text{Energy} \quad (17)$$

In this description, the binding energy is attributed to the field dynamics.

Black Holes and the Information Paradox

Black holes represent regions where the field becomes singular:

$$\lim_{r \rightarrow r_s} \delta m(r) \rightarrow \infty, \quad T(r) \rightarrow 0 \quad (18)$$

Information would be preserved in the field structure; this would be one possible approach to the information paradox.

.9 Philosophical Implications

From Particles to Excitation Patterns

The T0 theory suggests a view that moves from particle-based thinking to field patterns:

Viewpoint: From Particles to Patterns

Particle picture: Reality consists of separate particles interacting through forces

Field picture of T0 theory: Reality is excitation patterns in a universal field

Interpretation: What is fundamental would then be patterns and relationships rather than things

Unity in Diversity

In this picture, the diversity of particles appears as a unity expressing itself through different excitation modes:

$$\boxed{\text{One Field} \times \text{Infinite Patterns} = \text{All of Physics}} \quad (19)$$

The Question of Consciousness

If all matter reduces to field patterns, what about consciousness?

T0 perspective: Consciousness could be a self-referential pattern in the universal field — the field becoming aware of itself through localized excitation configurations.

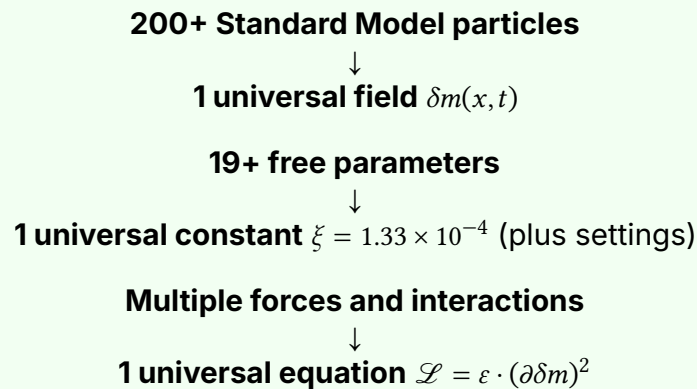
.10 Conclusion

Result

This analysis summarizes the unification approach of the T0 theory:

The Unification Approach

From a particle catalog to one field:



Aim: comparable predictive power with far fewer basic assumptions.

The Picture in Words

In the T0 description, the hundreds of different particles with their empirically fixed parameters are expressions of a single, universal field that manifests itself through an infinite spectrum of excitation patterns.

Every particle we have ever discovered — from the electron to the Higgs boson, from neutrinos to quarks — is, in this picture, a different form of excitation of the same field.

Context

The T0 theory pursues a unification idea that builds on Einstein's unification of space and time:

$$\text{Einstein: Space + Time} \rightarrow \text{Spacetime} \quad (20)$$

$$\text{T0 Theory: All Particles} \rightarrow \text{Universal Field} \quad (21)$$

The approach reads: one field, one equation, one parameter (plus motivated settings), an open spectrum of excitations.

The diversity of the particle spectrum may go back to a simpler common structure.

$$\boxed{\text{Reality} = \delta m(x, t) \text{ dancing the eternal patterns of existence}} \quad (22)$$

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